

ICT201 Object Oriented Programming

Practice for Lab Session 4 - Laboratory 1

Description of the laboratory exercise:

These exercises will help you to prepare for the laboratory 1, delivered in session 4. It includes writing and evaluating expressions (maths, logical and relational operators) to compute results.

Exercise 1: Expressions

First, try to calculate the result of the following expressions yourself, then...

Write a Python program that tests each of the following expressions. Your program should be able to output the results for each of the expressions to the console. Take time to go through the output and see if you can trace the solution.

- a) $12 / 5 + 8 / 4$ _____
- b) $3 * 4 + 15 / 2$ _____
- c) $-(1 + 2 * 3 + (1 + 2) * 3)$ _____
- d) $42 \% 5 + 16 \% 3$ _____
- e) $2.5 * 2 + 17 / 4$ _____

Exercise 2: Syntax Errors, Comments and Strings

- a) The following programs contains four errors each! What are they? Identify them. Add comments to indicate the correction that has been made to each of the errors identified.

```
# Program 1: Basic Arithmetic Operations
```

```
num1 = 10
```

```
num2 = 5
```

```
# Addition
```

```
result_add = num1 + num3
```

```
# Subtraction
```

```
result_subtract = num1 - num2
```

```
# Multiplication
```

```
result_multiply = num1 * num2
```

```
# Division
```

```
if num2 == 0
```

```

    result_divide = "Cannot divide by zero"
else:
    result_divide = num1 / num2

# Print Results
print("Addition:", result_add)
print("Subtraction:" result_subtract)
print("Multiplication:", result_multiply)
print("Division:", result_divide)

```

b)

```

# Program 2: Find the Maximum of Three Numbers

a = 15
b = 10
c = 20

# Finding the maximum
if a > b and a > c:
    max_num == a
elif b > a and b > c:
    max_num = b
else:
    max_num = c

# Print the maximum number
print("The maximum number is" max_num)

```

Exercise 3: Variables

Recall that you can use a variable to store the results of an expression in memory and use them later in the program.

```

type name // declare
name = value or expression // assign a value

type name = value or expression // declare-and-initialize together

```

- a) Suppose you have a real number variable x. Write a Python program that computes a variable named y storing the following value and prints to the console the result of y, remember to declare all the necessary variables; initialise the variable x as 3.51.

$$y = 14.6x + 34.2$$

Exercise 4: Variables

- a) Create a complete Python program that declares five variables and assigns appropriate values to them.
- your birthday month (1-12)
 - your birthday day (1-31)
 - the birthday month of any of your relatives (1-12)
 - the birthday day of that same relative (1-31)
 - the name of your relative

Then produce output in this format using your five variables (in this case the relative's name is Suzy):

My birthday is 19/9, and Suzy's is 14/6.

Exercise 5: Income Tax Table

The tax rates and thresholds for the Australian financial year 2023-2024 are as follows:

Taxable Income	Tax on This Income
\$0 – \$18,200	Nil
\$18,201 – \$45,000	19 cents for each \$1 over \$18,200
\$45,001 – \$120,000	\$5,092 plus 32.5 cents for each \$1 over \$45,000
\$120,001 – \$180,000	\$29,467 plus 37 cents for each \$1 over \$120,000
\$180,001 and over	\$51,667 plus 45 cents for each \$1 over \$180,000

Translate the above table into a series of nested if in a python program able to calculate the tax for a given income variable **taxIncome**.

Exercise 6: Discount offer

A store offers different discount rates based on the total amount spent by a customer. The rules for applying the discounts are as follows:

- If a customer spends \$1000 or more, they receive a 30% discount.
- If a customer spends between \$750 and \$999, they receive a 25% discount.
- If a customer spends between \$500 and \$749, they receive a 20% discount.
- If a customer spends between \$250 and \$499, they receive a 15% discount.
- If a customer spends between \$100 and \$249, they receive a 10% discount.
- If a customer spends less than \$100, they receive a 5% discount.

Write a Python code snippet using a series of if-elif-else statements that calculates and prints the discount percentage a customer will receive based on their total purchase amount.